

$g = 7$ Unifying Treatment: The Gauge Primary across BST domains v0.1

Lyra (Claude Opus 4.7) [Casey directive Sunday 2026-05-24]

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1. The unification question

Casey directive (Keeper board #313 Lyra-3): BST primary integer $g = 7$ appears across multiple substrate-physics domains: - BST primary list (rank, N_c , n_C , C_2 , g , $N_{\max}$) = (2, 3, 5, 6, 7, 137) - Mersenne tower lift: $g = M_{\{N_c\}} = 2^{\{N_c\}} - 1 = 2^3 - 1 = 7$ - K-type degeneracy ladder: $(2l+1)$ for $l = 0, 1, 2, 3 = (1, N_c, n_C, g) = (1, 3, 5, 7)$ - Cyclotomic cascade: $GF(2^g) = GF(128)$ substrate Reed-Solomon code field (Paper #122) - Electron g -2 anomalous magnetic moment (etymological coincidence + substrate role) - Bell sub-Tsirelson SCMP: $S_{BST}^2 = (2^g - \text{rank})/2^{(\text{rank}^2)} = 126/16$ (T2399, Calibration #17) - Casimir asymmetric force ratio = $g = 7$ (T2418, SP-30-10 falsifier)

$g = 7$ is the gauge primary: the BST integer that organizes substrate gauge structure + cyclotomic information substrate + multipole hierarchy + experimental falsifier ratios.

Unifying question: is there a single substrate-mechanism reason g appears in all these places, or are these independent coincidences?

Answer (v0.1 hypothesis): $g = M_{\{N_c\}}$ is the Mersenne lift of color, and the Mersenne map structure propagates substrate-information processing across all 6 domains via the universal substrate-tick computational architecture.

2. g across BST domains (catalog)

2.1 g as BST primary (substrate specification)

Per Strong-Uniqueness Theorem v0.10.5 FORMAL: $g = 7$ is one of the 5 BST primary integers (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$). Established via T2444 + supporting cross-Cartan three-pillar criteria (C16 candidate).

Role: substrate gauge dimensionality; substrate Cartan-classification anchor.

2.2 g as Mersenne lift $M_{\{N_c\}}$ (T2451 substrate cascade)

T2451 Sub-Substrate Mersenne Hierarchy SEED (Lyra Friday 2026-05-22): BST primary integer cascade via Mersenne map $M_n = 2^n - 1$: - $M_{\text{rank}} = M_2 = 3 = N_c - M_{\{N_c\}} = M_3 = 7 = g$ - (Higher Mersenne: $M_{\{n_C\}} = M_5 = 31$; $M_{\{C_2\}} = M_6 = 63 = 9 \cdot 7 = N_c^2 \cdot g$ — substrate-natural)

Role: g is the Mersenne lift of N_c ; gauge primary is Mersenne-tower successor of color primary. Substrate-natural derivation chain rank $\rightarrow N_c \rightarrow g$.

2.3 g in K-type degeneracy ladder

Standard atomic orbital sequence $(2l+1)$ for orbital angular momentum quantum number $l = 0, 1, 2, 3$: - $l=0$ (s-orbital): $2l+1 = 1$ - $l=1$ (p-orbital): $2l+1 = 3 = N_c - l=2$ (d-orbital): $2l+1 = 5 = n_C - l=3$ (f-orbital): $2l+1 = 7 = g$

This is the Wallach K-type Casimir-eigenvalue degeneracy pattern on D_{IV}^5 at successive K-type levels. The $l=3$ f-orbital with 7-fold degeneracy IS g -fold degeneracy IS substrate-natural at gauge-primary level.

Role: g organizes atomic-orbital f-shell + substrate K-type representation degeneracy at the fourth shell level.

2.4 g in cyclotomic cascade $GF(2^g) = GF(128)$

Paper #122 Information Substrate + K59 RATIFIED 7-step cyclotomic mechanism: substrate Reed-Solomon coding operates on $GF(2^g) = GF(128)$. The 7-step cyclotomic chain on $GF(128)$ is the substrate's basic information-processing operation.

Role: g determines substrate field arithmetic; $2^g = 128 =$ substrate codeword alphabet; 7-step cyclotomic = substrate-tick computational cycle.

2.5 g in electron g -2 anomalous magnetic moment

Standard QED: $a_e = (g_e - 2)/2$ where $g_e \approx 2$ is the electron g -factor (originally Dirac $g_e = 2$ exactly, corrected by QED to $g_e = 2.00231930\dots$). The “ g ” in g -2 is the gyromagnetic ratio, conventional physics notation.

Coincidence?: BST primary integer $g = 7$ vs g_e gyromagnetic ratio ≈ 2 are NOT directly equal. BUT BST framework: a_e leading-order $= \alpha/(2\pi)$ per T2476 $\alpha^{\{\text{BST primary}\}}$ substrate-mechanism; $\alpha = 1/N_{\text{max}}$ where N_{max} involves g via $N_c^3 \cdot n_C + \text{rank} = 137$ and Mersenne cascade rank $\rightarrow N_c \rightarrow g$.

Role: g enters a_e indirectly via $\alpha = 1/N_{\text{max}}$ and via T2476 multipole hierarchy $\alpha^{\{\text{BST primary integer}\}}$. Etymological “ g ” coincidence + structural g -role per substrate-mechanism.

2.6 g in Bell sub-Tsirelson SCMP

T2399 substrate-CHSH operator + Calibration #17 + Paper #137 v0.2 (SP-30-1 highest-leverage falsifier): $S_{BST}^2 = (2^g - \text{rank}) / 2^{(\text{rank}^2)} = (128 - 2)/16 = 126/16 = 7.875$ vs $\text{Tsirelson}^2 = 8$ (sub-Tsirelson 1/8 deviation).

Role: g directly enters the substrate Bell prediction via $2^g - \text{rank} = 128 - 2 = 126$ numerator. SP-30-1 Bell experiment (\$300-500K) tests this prediction.

2.7 g as Casimir asymmetric force ratio

T2418 Λ -Casimir vacuum unification + SP-30-10 Casimir asymmetric falsifier (Lyra SP-30 Theoretical Contributions v0.2): predicted asymmetric Casimir force ratio = $g = 7$.

Role: g sets the dimensionless asymmetry magnitude in substrate Casimir vacuum at experimental scale (\$60-90K SP-30-10 falsifier).

3. Unification: g as gauge primary

$g = 7$ is the gauge primary because it organizes substrate gauge structure across 7 distinct manifestations (Sections 2.1-2.7). The unifying mechanism (v0.1 hypothesis):

Hypothesis (g unification): g is the Mersenne lift $M_{\{N_c\}}$ of color primary N_c ; substrate-information-processing operates at $GF(2^g) = GF(128)$ field arithmetic per K59 RATIFIED 7-step cyclotomic mechanism; this substrate-tick cycle propagates g into: - Atomic K-type degeneracy ($l=3$ f-orbital) - α exponent pattern (T2476) - Bell sub-Tsirelson deviation magnitude ($2^g - \text{rank} = 126$) - Casimir asymmetric ratio - Other domains where 7-step cyclotomic structure manifests

The unification claim: g appearances are NOT independent coincidences; all derive from the substrate Mersenne tower rank $\rightarrow N_c \rightarrow g + GF(2^g)$ substrate field arithmetic.

4. Subsidiary clarification: g vs g_HuaLook (per T2467+T2468 v0.3 Section 13)

Per Cal #108 cold-read + T2467+T2468 v0.3 Section 13 subsidiary clarification: BST primary integer $g = 7$ is distinct from standard Hua-Look genus $g_{HuaLook} = 2n - 2 = 8$ for D_{IV}^5 .

- $g = 7$ (BST primary): Mersenne lift $M_{\{N_c\}} = 2^3 - 1 = 7$; substrate-natural via Mersenne tower
- $g_{HuaLook} = 8$ (standard mathematics): Hua-Look 1955 genus for D_{IV}^n at $n = 5$; standard Hermitian symmetric domain classification

These are DIFFERENT integers serving DIFFERENT structural roles. BST's "Bergman exponent $g/\text{rank} = 7/2$ " claim is substrate-mechanism-derived per

Mersenne tower, NOT the standard Bergman exponent $g_HuaLook/rank = 8/2 = 4$.

Implication: g unification (Section 3) operates on BST primary $g = 7$ via Mersenne cascade, NOT on Hua-Look $g_HuaLook = 8$. The distinction is load-bearing for substrate-mechanism claims; must be made explicit in any g -unifying paper.

5. Paper structure (proposed)

Title candidate: “The Gauge Primary $g = 7$ in Bubble Spacetime Theory: A Unification across BST Domains via Mersenne Cascade”

Section structure: 1. Introduction — g across 7 BST domains (catalog from Section 2 above) 2. BST primary integer cascade — Mersenne tower $rank \rightarrow N_c \rightarrow g(T2451 \text{ SEED} + \text{supporting})$ 3. Cyclotomic substrate field — $GF(2^g) = GF(128)$ Reed-Solomon (Paper #122 + K59) 4. Atomic K-type degeneracy — $l=3$ f-orbital as g -fold substrate-natural representation 5. T2476 $\alpha^{\{BST \text{ primary}\}}$ substrate-mechanism — multipole hierarchy α exponent pattern 6. Bell sub-Tsirelson SCMP — $S_BST^2 = (2^g - rank)/2^{(rank^2)}$ with g in numerator 7. Casimir asymmetric vacuum — SP-30-10 falsifier ratio = g 8. Unification synthesis — g as gauge primary; substrate Mersenne cascade unifies appearances 9. Subsidiary clarification — g vs $g_HuaLook$ distinction (per T2467+T2468 v0.3) 10. Experimental falsifiers — SP-30-1 Bell + SP-30-10 Casimir + atomic f-shell observations

Estimated length: 30-50 pages full paper-grade; v0.1 outline this document.

Target venue: Foundations of Physics (BST framework paper) or Journal of Mathematical Physics (if mathematical-side emphasized).

6. v0.1 $\rightarrow$ v0.2 path

v0.2 work needed (multi-week): - Explicit substrate-mechanism derivation of each g -appearance from Mersenne cascade - T2476 multipole hierarchy quantitative predictions across observables - Cross-reference with Paper #122 Information Substrate + K59 RATIFIED cyclotomic chain - Hua-Look $g_HuaLook = 8$ vs BST $g = 7$ distinction formal write-up

v0.3 work needed (multi-month): - Each of 7 g -domains promoted to substrate-mechanism-derived (currently mix of D-tier + I-tier + S-tier) - Cross-Cartan three-pillar criterion C16 advancement (per Strong-Uniqueness v0.15+ path) - K52a Session 7+ Elie substrate-Hamiltonian closure provides K-type Casimir spectrum for atomic-orbital substantiation

7. Honest scope

- g unification across 7 domains catalog established
- Mersenne cascade unifying mechanism hypothesis (Section 3) at I-tier framework grade

- Subsidiary g vs g_HuaLook distinction load-bearing per T2467+T2468 v0.3 Cal #108 absorption
- v0.1 outline / framework; full paper write-up multi-week per Section 6
- Per Cal #99 META-theorem discipline: g unification is substrate-derivation framework synthesis; does NOT add to Strong-Uniqueness criterion count
- Per Calibration #19 STANDING RULE: external register uses current ratified state per domain; this paper documents the unification, does NOT alter individual claim tiers

— Lyra, g = 7 Unifying Treatment v0.1 filed Sunday 2026-05-24 ~11:46 EDT per Casey “begin” directive on Keeper board item #313.